

TE Curve Verification Pipeline Directional Model Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report is the canonical TE Curve Verification Pipeline offline comparison between RCIM Model-Bank Reproduction , recovered original, retuned paper-reference model banks, and repository-owned Wave 1 and Wave 2.1 model candidates. It starts from the current direction-aware comparison matrix.

Dataset And Split

- dataset config: config/datasets/transmission_error_dataset.yaml ;
- dataset root: data\polished_dataset ;
- comparison mode: reduced_selected_model_matrix ;
- candidate count: 9 ;
- held-out curve count before candidate filtering: 94 ;
- percentage-error denominator: peak_to_peak_truth ;
- Fw candidates are evaluated only on forward curves;
- Bw candidates are evaluated only on backward curves;

Candidate Inventory

Candidate	Source	Family	Surface	Valid Directions
polished_rcim_model_bank_reproduction_GBM19_Bw	polished_rcim_model_bank_reproduction	GBM	Bw	backward
polished_rcim_model_bank_reproduction_ET19_Bw	polished_rcim_model_bank_reproduction	ET	Bw	backward
polished_feedforward_Bw	polished_model_development_registry	feedforward	Bw	backward
polished_tree_Bw	polished_model_development_registry	tree	Bw	backward
polished_harmonic_regression_Bw	polished_model_development_registry	harmonic_regression	Bw	backward
polished_periodic_gru_sequence_Bw	polished_model_development_registry	periodic_gru_sequence	Bw	backward
polished_periodic_mlp_harmonic_Bw	polished_model_development_registry	periodic_mlp_harmonic	Bw	backward
polished_wave4_3_mixture_density_k3_Bw	polished_model_development_registry	wave4_3_mixture_density_k3	Bw	backward
wave52b_offset_centered_shape_harmonic_Bw	wave52b_offset_harmonic_guided_registry	wave52b_offset_harmonic_guided	Bw	backward

Backward Comparison

Wave 5.2B Offset And Harmonic Guided Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_Bw	0.002266	0.002708	3.986	9.758

Polished Model-Development Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_periodic_gru_sequence_Bw	0.001129	0.001412	2.228	4.688
polished_wave4_3_mixture_density_k3_Bw	0.001930	0.002341	3.405	9.512
polished_periodic_mlp_harmonic_Bw	0.002396	0.002823	4.137	10.607
polished_feedforward_Bw	0.002655	0.003193	4.708	10.602
polished_tree_Bw	0.002756	0.003287	4.934	10.752
polished_harmonic_regression_Bw	0.004181	0.004742	8.213	15.543

Polished RCIM Model-Bank Reproduction Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_rcim_model_bank_reproduction_ET19_Bw	0.004034	0.004316	8.163	41.965
polished_rcim_model_bank_reproduction_GBM19_Bw	0.004482	0.004755	9.169	43.941

Curve Evidence

Each selected candidate is shown with the same four deterministic held-out operating conditions for this direction. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

Shared operating conditions

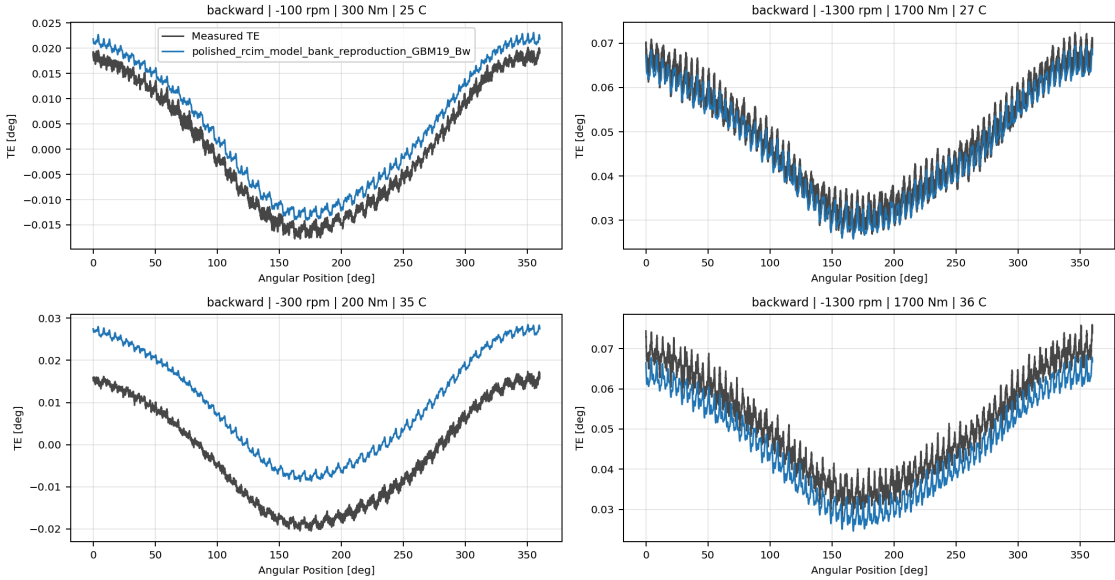
- 100 rpm, 300 Nm, 25 C
- 1300 rpm, 1700 Nm, 25 C
- 300 rpm, 200 Nm, 35 C
- 1300 rpm, 1700 Nm, 35 C

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
polished_rcim_model_bank_reproduction_GBM19_Bw	Selected candidate	0.004482	0.004755	9.169	43.941
polished_rcim_model_bank_reproduction_ET19_Bw	Selected candidate	0.004034	0.004316	8.163	41.965
polished_feedforward_Bw	Selected candidate	0.002655	0.003193	4.708	10.602
polished_tree_Bw	Selected candidate	0.002756	0.003287	4.934	10.752
polished_harmonic_regression_Bw	Selected candidate	0.004181	0.004742	8.213	15.543
polished_periodic_gru_sequence_Bw	Surface leader	0.001129	0.001412	2.228	4.688
polished_periodic_mlp_harmonic_Bw	Selected candidate	0.002396	0.002823	4.137	10.607
polished_wave4_3_mixture_density_k3_Bw	Selected candidate	0.001930	0.002341	3.405	9.512

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
wave52b_offset_centered_shape_harmonic_Bw	Selected candidate	0.002266	0.002708	3.986	9.758

polished_rcim_model_bank_reproduction_GBM19_Bw

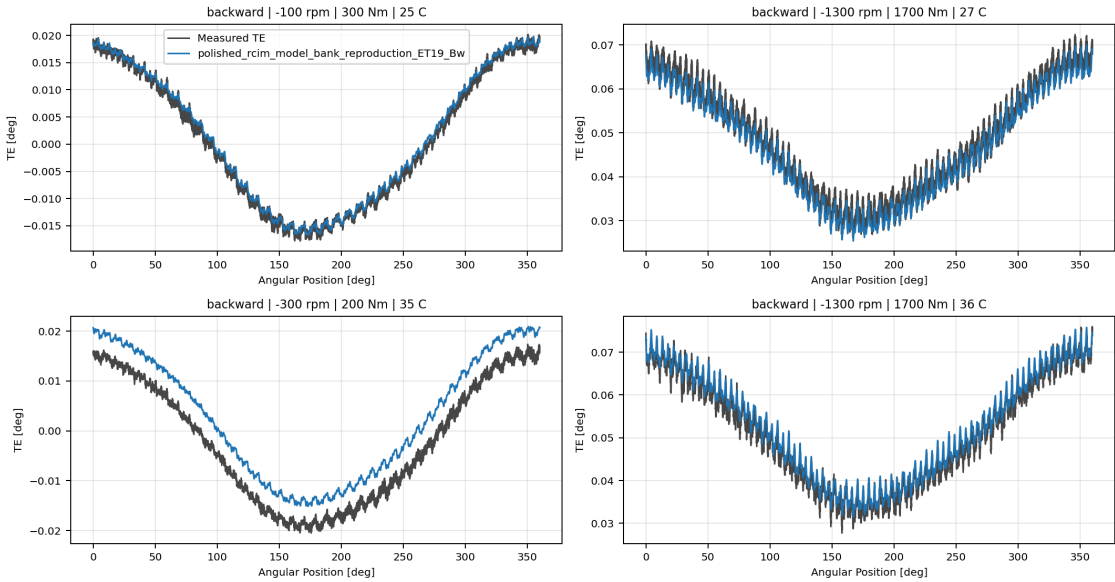
polished_rcim_model_bank_reproduction_GBM19_Bw



polished_rcim_model_bank_reproduction_GBM19_Bw measured-versus-predicted TE collage

polished_rcim_model_bank_reproduction_ET19_Bw

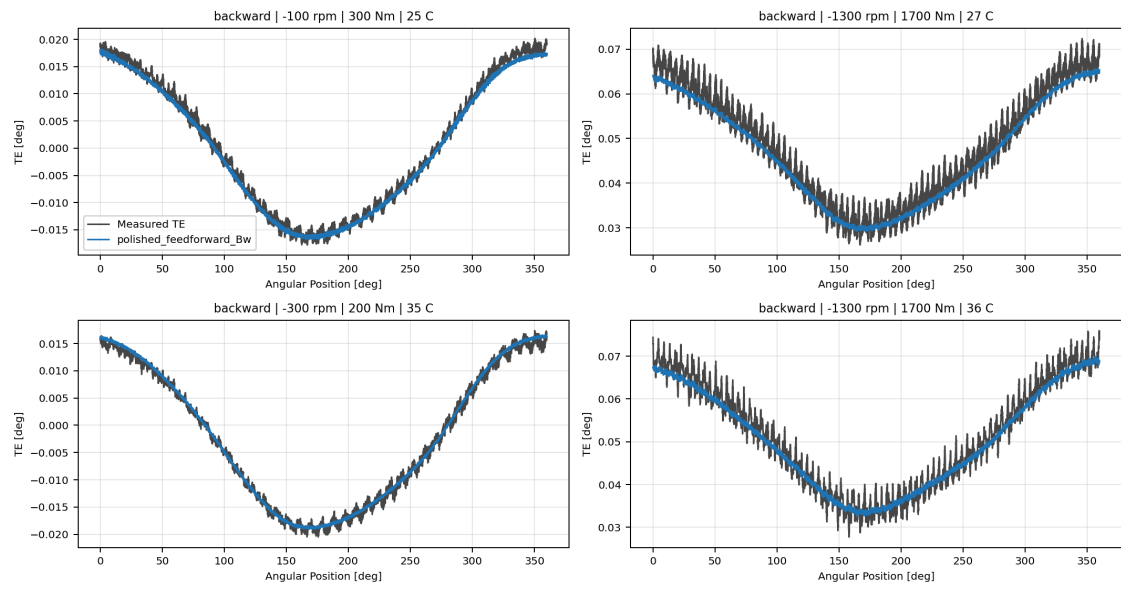
polished_rcim_model_bank_reproduction_ET19_Bw



polished_rcim_model_bank_reproduction_ET19_Bw measured-versus-predicted TE collage

polished_feedforward_Bw

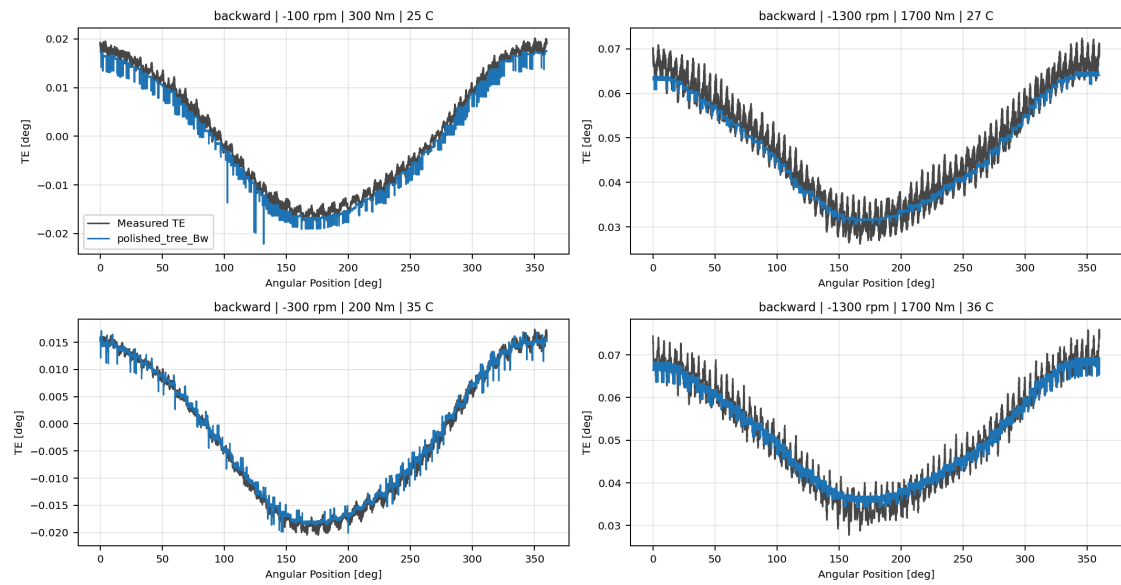
polished_feedforward_Bw



polished_feedforward_Bw measured-versus-predicted TE collage

polished_tree_Bw

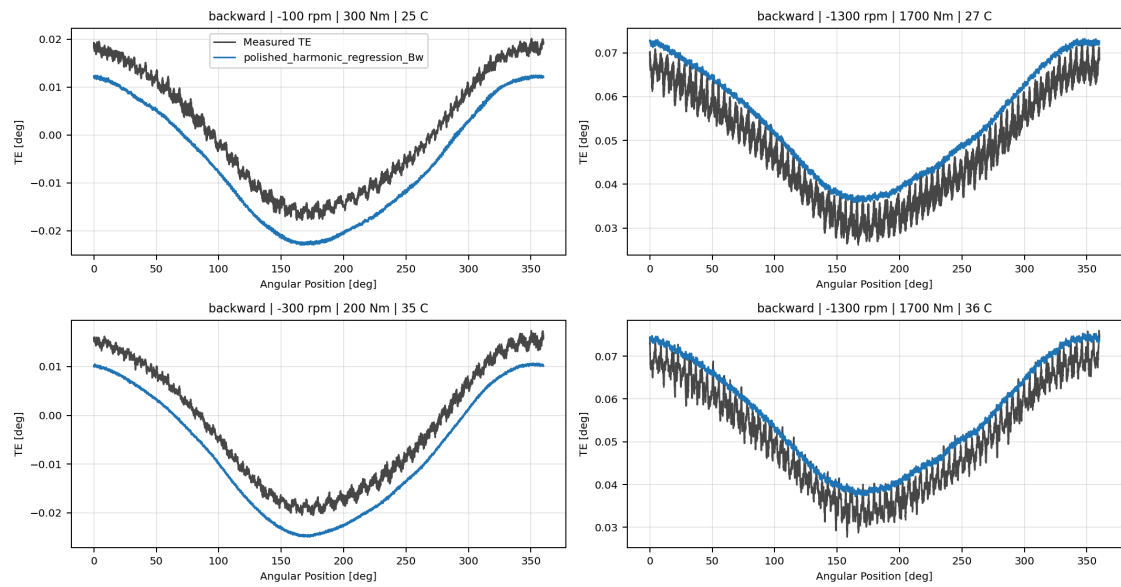
polished_tree_Bw



polished_tree_Bw measured-versus-predicted TE collage

polished_harmonic_regression_Bw

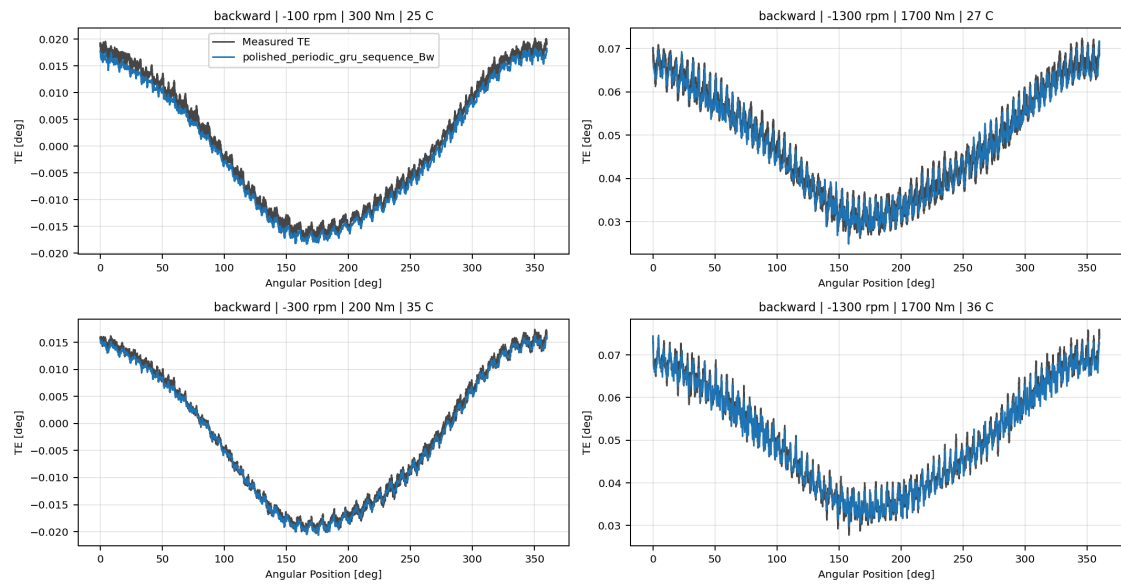
polished_harmonic_regression_Bw



polished_harmonic_regression_Bw measured-versus-predicted TE collage

polished_periodic_gru_sequence_Bw

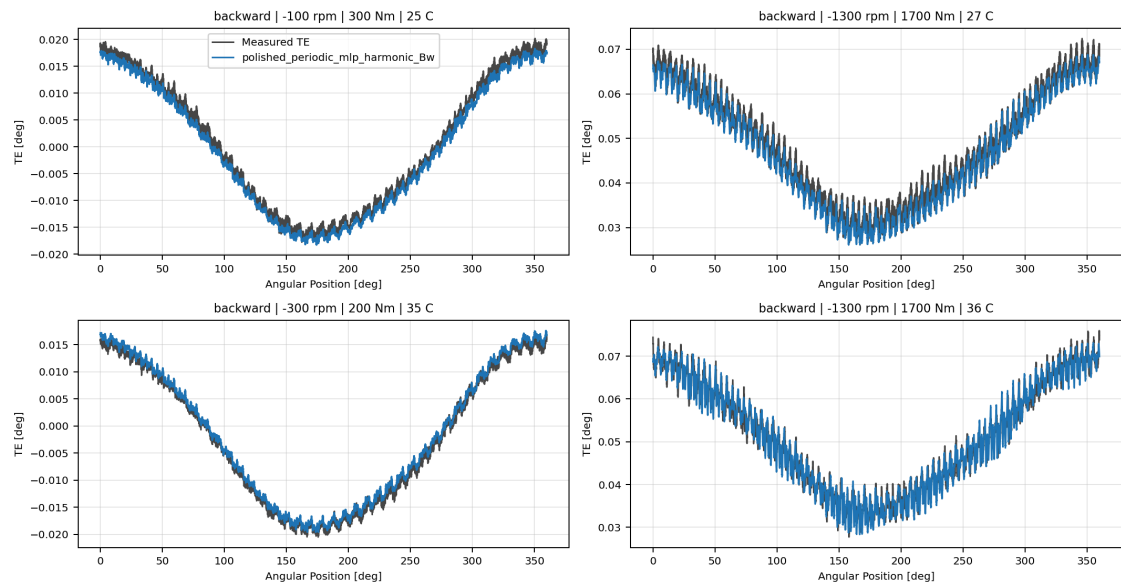
polished_periodic_gru_sequence_Bw



polished_periodic_gru_sequence_Bw measured-versus-predicted TE collage

polished_periodic_mlp_harmonic_Bw

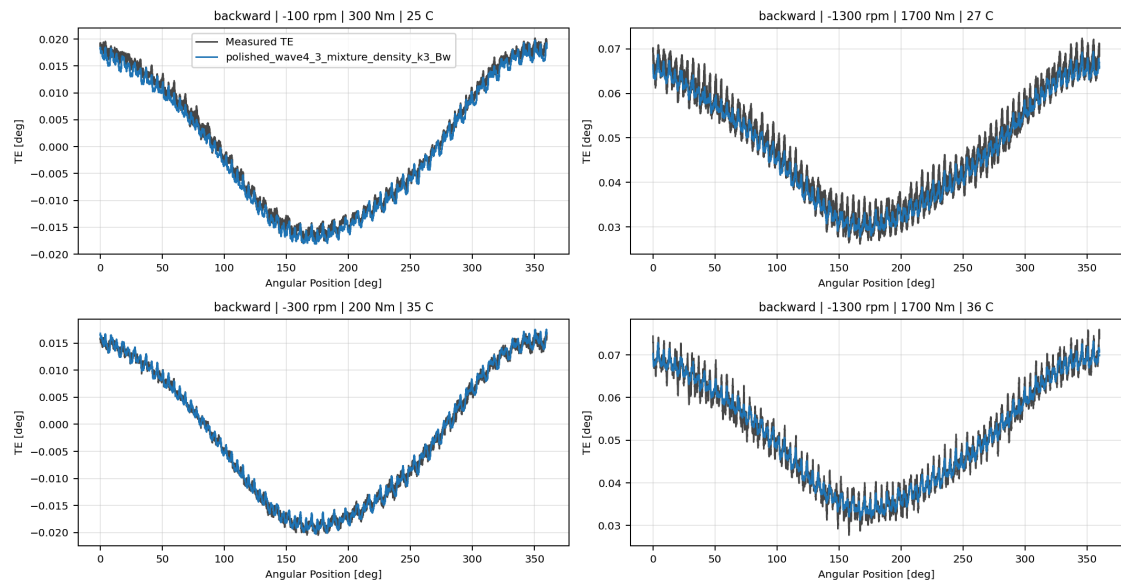
polished_periodic_mlp_harmonic_Bw



polished_periodic_mlp_harmonic_Bw measured-versus-predicted TE collage

polished_wave4_3_mixture_density_k3_Bw

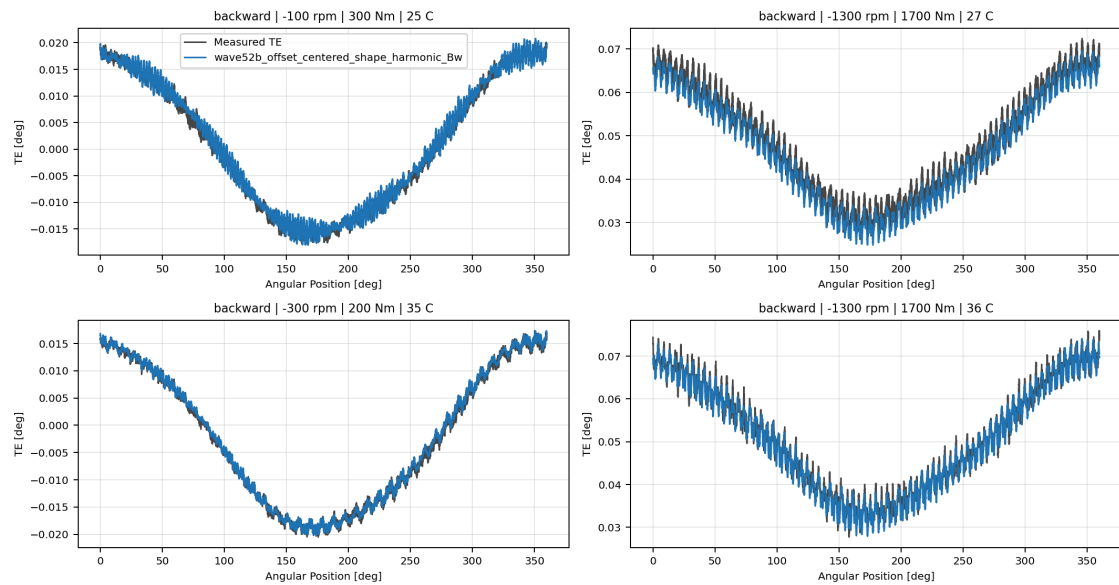
polished_wave4_3_mixture_density_k3_Bw



polished_wave4_3_mixture_density_k3_Bw measured-versus-predicted TE collage

wave52b_offset_centered_shape_harmonic_Bw

wave52b_offset_centered_shape_harmonic_Bw



wave52b_offset_centered_shape_harmonic_Bw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-07-01-18-57_track2_reduced_selected_model_matrix_track2_selected_polished_dataset_backward_2026_07_06\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-07-01-18-57_track2_reduced_selected_model_matrix_track2_selected_polished_dataset_backward_2026_07_06\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\paused_reduced_selected_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within each source group and direction. Directional paper-reference, Wave 1, and Wave 2.1 models are never evaluated on the opposite direction. Global Wave models remain valid on both directions and are therefore shown in the directional sections and again in the global breakdown. The `rcim_track1` forward reference banks use the opposite stored `h0` sign convention relative to the TE Curve Verification Pipeline reconstruction contract, so the TE Curve Verification Pipeline comparison applies the documented source-specific `h0` compatibility multiplier before curve reconstruction.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online Table 9 compensation benchmark.
- The report uses the saved Python model artifacts from `models/` ; ONNX parity checks remain a separate deployment-readiness task.